COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE-010: Data Structures and Algorithm	**SECTION:**	CPE21S3
DATE PERFORMED:	7/9/26	**DATE SUBMITTED:**	7/14/26
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ACTIVITY NO. 1.1 **Basic C++ Programming**

1. PROCEDURE OUTPUT

ILO A:

```cpp
//Header File Declaration Section
#include <iostream>

//Global Declaration Section
struct var{
    int num1, num2;
    bool a, b;
};

//Class Declaration and Method Definition Section
void sum(var v){
    std::cout << "Sum is " << v.num1 + v.num2 << std::endl;
}

void compare(var v){
    if(v.num1 > v.num2){
        std::cout << v.num1 << " is greater than " << v.num2 << std::endl;
    }
    else if(v.num1 < v.num2){
        std::cout << v.num1 << " is less than " << v.num2 << std::endl;
    }
    else{
        std::cout << v.num1 << " is equal to " << v.num2 << std::endl;
    }
}

void boolean(var v){
    bool and_gate = v.a && v.b;
    bool or_gate = v.a || v.b;

    std::cout << "Check for and: " << and_gate << std::endl;
    std::cout << "Check for or: " << or_gate << std::endl;
}
```

```cpp
//Main Function
int main(){
    var v;
    int choice = 0;

//Method Definition
    while(choice != 4){
        std::cout << "\n1. Sum\n2. Compare\n3. Boolean\n4. Exit\nChoice: ";
        std::cin >> choice;

        switch(choice){
            case 1:
                std::cout << "Enter two numbers: ";
                std::cin >> v.num1 >> v.num2;
                sum(v);
                break;

            case 2:
                std::cout << "Enter two numbers: ";
                std::cin >> v.num1 >> v.num2;
                compare(v);
                break;

            case 3:
                std::cout << "Enter two Boolean values 0 or 1: ";
                std::cin >> v.a >> v.b;
                boolean(v);
                break;

            case 4:
                std::cout << "Program ended.";
                break;

             default:
                 std::cout << "Invalid choice.";
        }
    }
    return 0;
}
```

```
1. Sum
2. Compare
3. Boolean
4. Exit
Choice: 1
Enter two numbers: 2
2
Sum is 4

1. Sum
2. Compare
3. Boolean
4. Exit
Choice: 2
Enter two numbers: 4
5
4 is less than 5

1. Sum
2. Compare
3. Boolean
4. Exit
Choice: 2
Enter two numbers: 5
3
5 is greater than 3

1. Sum
2. Compare
3. Boolean
4. Exit
Choice: 10 10
Invalid choice.
1. Sum
2. Compare
3. Boolean
4. Exit
```

```
Choice: Invalid choice.
1. Sum
2. Compare
3. Boolean
4. Exit
Choice: 2
Enter two numbers: 10 10
10 is equal to 10

1. Sum
2. Compare
3. Boolean
4. Exit
Choice: 3
Enter two Boolean values 0 or 1: 0 1
Check for and: 0
Check for or: 1

1. Sum
2. Compare
3. Boolean
4. Exit
Choice: 3
Enter two Boolean values 0 or 1: 1 1
Check for and: 1
Check for or: 1

1. Sum
2. Compare
3. Boolean
4. Exit
Choice: 3
Enter two Boolean values 0 or 1: 0 0
Check for and: 0
Check for or: 0
```

```
1. Sum
2. Compare
3. Boolean
4. Exit
Choice: 4
Program ended.
--------------------------------
Process exited after 313.6 seconds with return v
alue 0
Press any key to continue . . .
```

ILO B:

```cpp
#include<iostream>

// naming convention or class name: PascalCase
class Triangle{
    private:
        double totalAngle, angleA, angleB, angleC;
    public:
        Triangle(double A, double B, double C);
        void setAngles(double A, double B, double C);
        const bool validateTriangle();
};

int main() {
    Triangle set1(40,30,100);
    if(set1.validateTriangle()){
        std::cout<<"The shape is a valid triangle\n";
    }
    else{
        std::cout<<"The shape is NOT a valid triangle\n";
    }
    return 0;
}

Triangle::Triangle(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}

void Triangle::setAngles(double A, double B, double C){
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}

const bool Triangle::validateTriangle(){
    return (totalAngle <= 180);
}
```

```
C:\Users\TIPQC\Documents\S
The shape is a valid triangle

--------------------------------
Process exited after 0.009917 seconds with return value 0
Press any key to continue . . .
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C:

```
#include <iostream>
#include <cmath>

struct var{
    double num1, num2;
    double kelvin;
    double x1, y1, x2, y2;
};

void swapping(var v){
    double temp = v.num1;
    v.num1 = v.num2;
    v.num2 = temp;

    std::cout << "Both swapped places: " << v.num1 << " " << v.num2 << std::endl;
}

void ktof(var v){
    std::cout << "Fahrenheit: " << (v.kelvin - 273.15) * 9 / 5 + 32 << std::endl;
}

void distance(var v){
    double d = sqrt((v.x2 - v.x1) * (v.x2 - v.x1) +
                    (v.y2 - v.y1) * (v.y2 - v.y1));

    std::cout << "The distance is " << d << std::endl;
}
```

```
int main(){
    var v;
    int choice = 0;

    while(choice != 4){
        std::cout << "\n1. Swap Numbers\n2. Kelvin to Fahrenheit\n3. Distance Between Points\n4. Exit\nInput: "
        std::cin >> choice;
        std::cout << "\nInput numbers: ";

        switch(choice){
            case 1:
                std::cin >> v.num1 >> v.num2;
                swapping(v);
                break;

            case 2:
                std::cin >> v.kelvin;
                ktof(v);
                break;

            case 3:
                std::cin >> v.x1 >> v.y1 >> v.x2 >> v.y2;
                distance(v);
                break;

            case 4:
                std::cout << "Program ended.";
                break;

            default:
                std::cout << "Invalid choice.";
        }
    }
    return 0;
}
```

```
1. Swap Numbers
2. Kelvin to Fahrenheit
3. Distance Between Points
4. Exit
Input: 1

Input numbers: 34 49
Both swapped places: 49 34

1. Swap Numbers
2. Kelvin to Fahrenheit
3. Distance Between Points
4. Exit
Input: 2

Input numbers: 100
Fahrenheit: -279.67

1. Swap Numbers
2. Kelvin to Fahrenheit
3. Distance Between Points
4. Exit
Input: 3

Input numbers: 3 4 6 8
The distance is 5
```

```
1. Swap Numbers
2. Kelvin to Fahrenheit
3. Distance Between Points
4. Exit
Input: 4

Input numbers: Program ended.
--------------------------------
Process exited after 27.12 seconds with return v
alue 0
Press any key to continue . . .
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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